

DUBUQUE, IA, USA

WMO: 725470

Lat: 42.398N

Lon: 90.704W

Elev: 322

StdP: 97.52

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.6	-19.6	-26.3	0.4	-21.9	-23.7	0.5	-19.2	12.2	-6.7	11.3	-5.3	4.0	290	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.1	(c) 31.3	(d) 23.9	(e) 29.8	(f) 23.0	(g) 28.4	(h) 22.0	(i) 25.3	(j) 29.5	(k) 24.2	(l) 28.1	(m) 23.3	(n) 27.0	(o) 5.2	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.7	28.0	23.0	18.4	27.0	22.0	17.3	25.8	79.2	29.5	74.8	28.1	70.8	27.0	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.5	(b) 10.1	(c) 8.9	(e) -26.8	(f) 33.4	(g) 3.7	(h) 1.7	(i) -29.5	(j) 34.6	(k) -31.7	(l) 35.6	(m) -33.8	(n) 36.6	(o) -36.5	(p) 37.8		
(a) 11.5	(b) 10.1	(c) 8.9	(e) -27.0	(f) 27.0	(g) 3.5	(h) 0.9	(i) -29.5	(j) 27.7	(k) -31.5	(l) 28.3	(m) -33.5	(n) 28.8	(o) -36.0	(p) 29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-7.3	-5.0	1.9	8.8	14.9	20.3	22.2	21.1	17.2	10.2	2.7	-4.1
	DBStd	11.56	6.97	6.71	6.49	5.23	4.60	3.47	3.03	2.87	4.34	5.17	5.63	6.44
	HDD10.0	2036	536	419	263	82	9	0	0	0	3	61	226	437
	HDD18.3	3892	794	653	510	288	125	19	4	8	72	256	468	694
	CDD10.0	1549	0	1	12	46	163	309	378	345	219	68	8	1
	CDD18.3	363	0	0	1	2	20	79	124	95	38	4	0	0
	CDH23.3	2831	0	0	3	28	167	626	1023	660	287	37	0	0
	CDH26.7	751	0	0	0	3	34	169	312	160	66	6	0	0
Wind	WSAvg	4.4	4.9	5.0	5.0	5.3	4.6	3.8	3.3	3.1	3.6	4.4	4.8	4.7
Precipitation	PrecAvg	961	33	36	70	96	110	111	99	117	113	67	69	46
	PrecMax	1610	153	79	151	195	240	266	311	252	393	218	270	105
	PrecMin	573	8	6	10	21	18	18	22	2	2	0	9	4
	PrecStd	218	26	21	39	43	50	66	64	69	90	45	52	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	13.8	22.8	26.7	30.1	32.3	34.0	32.6	30.8	27.4	19.9	14.3
		MCWB	6.6	9.6	16.3	16.7	21.2	23.9	25.5	24.1	23.1	19.8	13.6	13.3
	2%	DB	5.5	9.0	18.1	23.2	27.3	30.4	31.4	30.1	28.6	23.6	16.0	9.6
		MCWB	3.6	6.4	12.9	15.2	19.5	22.9	24.8	23.5	21.4	17.3	11.9	7.8
	5%	DB	3.2	5.8	14.5	20.3	25.1	28.7	29.8	28.4	26.7	21.1	13.5	6.3
		MCWB	1.8	3.7	10.8	13.3	18.0	21.7	23.6	22.6	20.0	16.3	10.4	4.5
	10%	DB	1.5	3.4	11.0	17.7	23.0	27.0	28.3	26.9	24.6	18.5	11.0	3.7
		MCWB	0.5	1.8	7.4	11.7	17.1	20.7	22.5	21.6	19.1	14.2	8.0	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	10.4	16.9	18.5	22.7	25.5	27.5	26.1	24.3	21.3	15.1	13.5
		MCDB	7.9	12.9	22.1	23.9	28.3	30.4	31.9	30.5	28.8	25.4	18.3	14.5
	2%	WB	3.9	6.7	14.2	16.3	21.0	24.0	25.7	24.8	22.8	19.1	12.9	7.9
		MCDB	5.5	9.0	17.0	21.7	25.5	28.6	29.9	28.5	26.7	21.9	15.4	8.9
	5%	WB	2.0	3.9	10.9	14.6	19.5	23.0	24.7	23.5	21.5	17.0	10.8	4.7
		MCDB	3.0	5.6	14.3	19.5	23.6	27.4	28.6	26.9	24.9	20.0	13.1	5.9
	10%	WB	0.5	1.9	7.8	12.4	18.1	21.8	23.7	22.6	20.2	15.0	8.2	2.4
		MCDB	1.5	3.2	10.9	16.9	21.9	25.9	27.2	25.8	23.4	18.3	10.7	3.6
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.2	9.3	10.7	10.7	10.1	10.1	10.0	11.0	10.2	8.7	7.5
		MCDBR	9.3	10.0	13.3	14.9	12.8	11.6	11.1	11.0	12.7	13.0	12.1	9.8
		MCWBR	7.7	7.7	8.5	8.7	6.9	6.2	5.7	5.8	6.5	7.7	8.7	8.0
	5% WB	MCDBR	8.9	9.7	12.3	13.1	11.1	10.5	10.1	9.7	10.6	11.2	11.2	9.0
		MCWBR	7.5	7.8	8.6	8.6	7.0	6.2	5.8	5.8	6.3	8.1	9.0	8.0
Clear-Sky Solar Irradiance	taub	0.280	0.297	0.333	0.371	0.400	0.419	0.430	0.408	0.379	0.335	0.292	0.278	
	taud	2.396	2.354	2.378	2.294	2.273	2.259	2.222	2.285	2.340	2.465	2.512	2.434	
	Ebn at Noon	868	907	911	896	875	856	841	849	850	854	853	840	
	Edh at Noon	78	96	107	126	132	134	138	126	110	86	70	69	
All-Sky Solar Radiation	RadAvg	1.77	2.77	3.76	4.63	5.47	6.10	6.09	5.35	4.46	2.90	1.92	1.40	
	RadStd	0.21	0.30	0.38	0.38	0.39	0.50	0.39	0.36	0.35	0.36	0.15	0.12	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page